

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EE-7101 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Computer Aided Design of Electrical Machines
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Give designing steps for D.C. motor 3 point starter.
2. Explain the working of a 4 point DC motor starter.
3. What is CAD? Explain its importance in design of electrical equipment.
4. a) Describe an electromagnet? What are some common devices that contain electromagnets?
b) Describe the role of the electromagnet in a doorbell.
5. What is the working principal of a choke coil? What is the function of a choke in tube light?
6. Discuss briefly the changes you would suggest in the design of a three phase squirrel cage induction motor to achieve the following:
 - i) Increase starting torque
 - ii) Increase full load power factor
 - iii) Reduce slip
 - iv) Increase efficiency
7. Discuss the design in respect of single phase induction motor.
 - i) Calculation main dimension of stator.
 - ii) Design of main and auxiliary winding.
8. Write a short notes (Any two)
 - i) Design of slip ring and squirrel cage rotor
 - ii) Design consideration of turbo and water wheel alternators.
 - iii) Optimization of design constrained and unconstrained optimization problem.
 - iv) Design of field systems of synchronous machine
